AI-Driven Emergency Care Coordination — System Blueprint (v0.2)

Date: 2025-08-25
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Scope: Urban ED (adult), advice-only assistant with human-in-the-loop (HITL), event-sourced orchestration.

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0) Objectives

- Reduce coordination latency from EMS pre-alert → specialist handoff while preserving clinical control.
- Offload admin/cognitive burden with automation of tracking, notifications, and pre-filled orders that *require approval*.
- Maintain rigorous auditability, safety guardrails, and GDPR compliance.

Primary KPIs
- Minutes saved per patient (coordination time).
- Time-to-treatment for urgent cases (e.g., stroke/trauma/sepsis).
- Fewer interruptions/pages per clinician per shift.
- Reliability of information flow (handoff completeness, miss rate of follow-ups).

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1) System Invariants (must hold)

- Advice-only: model(s) *never* place orders/transfer autonomously; every action produces a proposal that requires explicit approval.
- Gate-first: rule/gate engine remains the arbiter; learning components only refine/prioritize suggestions.
- Frozen core interfaces: WorkflowState, TinyCritics, CSV/dir semantics for tracker/SOP/QR, lazy optional deps; UI thresholds for overdue logic.
- All new features behind runtime flags (e.g., `RUN_PIPELINE`, `RUN_UI`).
- Full append-only audit with immutable linking (hash-chained records) and export for external verification.

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2) Target Architecture (event-sourced, modular)

Pattern: Event Bus (Pub/Sub) + Micro-services. Each service is idempotent; commands and queries are separated.

Core services 1. **Ingest** — EMS/HL7/JSON input → canonical events.
2. **Gate Engine** — deterministic safety & policy gates (ICU capacity, BGA/critical labs, duplication checks).

3. **Action Heads** — multi-head proposal generator (e.g., `Order:CT`, `Order:Labs`, `Notify:Neuro`, `Bed:ICU`) using simple baselines now, ML refine-only later.
4. **Calibration & Ranking** — per-action thresholding, temperature scaling, cost-aware ranking.
5. **Policy Layer (HITL)** — human approval rules, staleness checks, escalation policies.
6. **Bridge/Adapters** — map approved actions to HL7/ORM/ORU or hospital JSON APIs.
7. **Capacity Service** — real-time/near-real-time bed/ICU/CT slot view with freshness TTLs.
8. **Tracker & SOPs** — equipment movement/status, SOP registry (offline-safe).
9. **Notifier** — targeted alerts (pager/SMS/ED dashboard) with dedup suppression & quiet hours.
10. **Audit & Telemetry** — append-only event store, metrics sink, and nightly KPI jobs.

Hot path
`Ingest → Gate Engine → Action Heads → Calibration → Policy (HITL) → Bridge → Notifier`
Audit and metrics run *in parallel*, never blocking the hot path.

Data flow
- **Event log** is the source of truth (event-sourcing).
- Services subscribe to topics: `EMS.PREALERT`, `ED.ARRIVAL`, `ORDER.PLACED`, `RESULT.READY`, `BED.REQUESTED`, `CONSULT.ACK`, etc.
- State is reconstructed via `WorkflowState.apply(event*)` when needed.

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3) Workflow Mapping → Modules & Topics

1) **EMS On Scene**
- **Input:** structured triage (trauma/non-trauma, vitals, allergies, past admissions).
- **Topics:** `EMS.PREALERT`, `EMS.VITALS`.
- **Actions proposed:** pre-alert neuro/trauma; prepare CT/labs; fetch prior admissions.

2) **Capacity Check**
- **Service:** Capacity with freshness TTL, fallbacks when stale.
- **Actions proposed:** `Bed:ICU tentative`, `CT:slot hold` (soft hold tokens), or deferral/escalation.

3) **ED Arrival & Reassessment**
- **Input:** reassessment vitals; allergy/safety checks; reconciliation with EMS data.
- **Actions proposed:** pre-filled order sets (CT, labs, ECG) → **Policy** requires human approval.

4) **Intra-ED Coordination**
- **Tracking:** diagnostics lifecycle; result watchers with SLA timers (e.g., troponin delta due in 60–90 min).
- **Actions proposed:** `Notify:on result`, `Order:follow-up` reminders, `Nudge:overdue equipment/vitals`.

5) **Specialist Handoff**
- **Proposal:** best receiving department based on trajectory (advice-only) + transfer orchestration.
- **Bridge:** emit standardized handoff JSON/HL7 + QR shortlink to full context.

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4) Data Contracts (canonical JSON)

4.1 Event envelope

```
{
  "event_id": "uuid",
  "ts": "2025-08-25T12:34:56Z",
  "patient_id": "anon-123",
  "type": "EMS.PREALERT",
  "payload": { /* type-specific */ },
  "provenance": {"source": "ems-device-42", "operator": "ems-crew-A"}
}
```

4.2 Key payloads (examples)

- **Vitals**: HR, RR, SpO2, SBP/DBP, Temp, GCS; derivations for qSOFA/MEWS.
- **Labs/HL7 map**: canonical keys (e.g., `vbg-pH`, `vbg-lactate`, `lab-ddimer-mgL-FEU`, `troponin0-ngL`).
- **Capacity**: `icu_free`, `ct_queue_len`, `last_updated`, `ttl_sec`.

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5) Gate Engine (deterministic)

- ICU capacity thresholds: `blocked ≤ 0`, `tight 0–15%`, `ok ≥ 15%`.
- BGA/critical ranges for pH, K+, Na+, lactate.
- Staleness: deny proposals if capacity/critical context is stale beyond TTL.
- Dedup & conflict: prevent duplicate orders; merge gates with TTL = **max** across packs; priority = **max**.

Outputs: structured `GatePack` with `alerts`, `blocks`, `next_action_hint` (never auto-exec).

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6) Learning Components (refine-only)

- **Backbone Port:** `(B,T,F) -> (B,H)`; initial backbone = GRU.
- **Action Heads:** multi-head (one head per action family).
- **Calibration:** per-head temp scaling; threshold sweep with cost matrix (false alarm vs. miss).
- **Never veto safety gates**; can reorder/prioritize within allowed set.

Advice outputs schema

```
{
  "action": "Order:CT",
  "params": {"protocol": "non-contrast head"},
```

```
  "score": 0.83,
  "explain": ["recent neuro deficit", "SBP>180", "last-labs<stale"]
}
```

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7) Policy Layer & HITL UX

- Approval states: `PROPOSED → APPROVED/REJECTED/EXPLAINED` (explain requires note).
- Guardrails:
- No order without explicit approval.
- “Two-man rule” optional for high-risk actions.
- Capacity staleness or conflicts → block with rationale.
- UI affordances: approve/deny with one click, inline rationale, QR to patient context, adjustable overdue thresholds (minutes).

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8) Bridge/Integrations

- HL7/ORM/ORU emitters; hospital JSON adapters.
- Idempotent delivery with retry & backoff; at-least-once semantics; dedup keys `(patient_id, action, timestamp)`.

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9) Audit, Security, Compliance

- **Audit**: append-only log, hash-chained (Merkle root per batch).
- **Anchoring**: periodic root anchoring to external ledger (permissioned/public) — **off hot path**.
- **Security**: mTLS between services; KMS-managed signing keys; role-based approvals; redaction at capture.
- **GDPR**: no PHI in anchors; erasure supported off-chain; logs retain only hashes/metadata.

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10) Telemetry & KPI Measurement

- Per action: PR-AUC, F1@τ, alarm fatigue rate, latency (egress time − ingress time).
- Per patient: minutes saved (baseline vs. assisted), number of approvals, #interruptions avoided.
- System SLOs: 99p hot-path latency, event ingestion availability, staleness rate of capacity data.

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11) Test & Replay Harness

- Deterministic replays from event logs; seed-fixed models; shadow mode support.
- Chaos tests: capacity stale, adapter down, duplicate events, delayed lab results.
- Delivery checklist mirroring invariants; evidence = smoke writes + file presence.

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12) API Sketch (REST + Topics)

REST (FastAPI sketch) - `POST /ems/prealert` → emit `EMS.PREALERT`
- `POST /ed/arrival` → emit `ED.ARRIVAL`
- `POST /capacity/update` → upsert capacity feed
- `POST /action/propose` (internal)
- `POST /action/{id}/approve|reject`
- `GET /patient/{id}/state`
- `GET /audit/{id}`

Topics
`EMS.*`, `ED.*`, `ORDER.*`, `RESULT.*`, `BED.*`, `CONSULT.*`, `NOTIFY.*`

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13) MVP Scope (2–3 weeks realistic)

M0 (Days 1–3)
- Event Bus (in-proc or lightweight broker), canonical events, Gate Engine baseline, Capacity Service with TTL.
- Append-only audit w/ hash-chain; nightly anchor placeholder.

M1 (Days 4–10)
- Multi-Head Action Heads (rules/baseline), Calibration module, Policy/HITL UI (Streamlit) with adjustable thresholds.
- Bridge stubs (HL7/JSON) with idempotent delivery.
- Result watchers (lab/CT) + reminders.

M2 (Days 11–15)
- Specialist handoff suggester (rule-based) + QR link.
- Shadow-mode ML refine-only path (GRU backbone port wired, no auto-exec).
- Replay harness + KPI dashboard (minutes saved, alarms).

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14) Risks & Mitigations

- **Data staleness (capacity/ICU)** → TTL + block on stale; visualize freshness.
- **Alarm fatigue** → per-action calibration + cost-aware thresholds; dedup in Notifier.
- **Integration fragility** → adapters idempotent, retries, dead-letter queues.
- **Scope creep toward autonomy** → enforce HITL policy tests in CI; approvals required.
- **Security** → mTLS, KMS, principle of least privilege, secrets rotation.

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15) Open Assumptions (to be validated without blocking build)

- Adult ED focus; pediatric flows out of scope for v0.2.
- Hospital exposes HL7 or JSON endpoints for orders/transfers.
- EMS pre-alert can deliver structured fields listed above.
- Local deployment possible (on-prem or VPC) with access to internal systems.

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16) Definition of Done (v0.2)

- End-to-end demo: EMS pre-alert → ED arrival → proposed orders → approved → bridge emits → notifications; audit & KPIs recorded.
- Contract & invariants upheld; smoke tests pass; evidence artifacts produced.
- Shadow-mode ML path wired (off by default), no autonomy, reproducible replay.